Supplementary Material: Proofs for "Learnable and Transferable Logical Operators for Complex Query Answering on Knowledge Graph Foundation Models"

I. Notation and Conventions

A knowledge graph is a finite structure $G = (V, R, E)$ with $E \subseteq V \times R \times V$. G^+ denotes the closure under inverse relations $\bar{r}$ (one fresh symbol per $r \in R$); all queries are evaluated over G^+ so that atom directions are without loss of generality. For a pattern (witness) structure W with a distinguished anchor node η and $e \in V$, $\hom_W(G, e)$ is the number of homomorphisms $W \to G^+$ with $\eta \mapsto e$. Queries follow Definition 1 of the main text (EPFO with atom-level negation, DNF disjunction); $\mathbf{1}[\cdot]$ is the indicator.

II. Proof of Theorem 1 (Canonical Representation)

Theorem 1. *For every EPFO$^\neg$ query q there is a finite family $\mathcal{W}_q$ of anchored witness structures such that for every G and $e \in V(G)$, $\mathbf{1}[e \in q(G)]$ is an integer combination of $\{\hom_W(G, e)\}_{W \in \mathcal{W}_q}$; negated atoms enter through inclusion–exclusion.*

Proof. Step 1 (positive conjunctive core). Let q° be the positive conjunctive part with variables $x_1, \ldots, x_m$ (including the answer variable $x_?$). Associated with q° is its *canonical database* $D(q^\circ)$: the typed pattern graph whose nodes are the variables and whose typed edges are the atoms. By the classical query-to-homomorphism correspondence, $e \in q^\circ(G) \iff \exists$ homomorphism $h : D(q^\circ) \to G^+$ with $h(x_?) = e$. Partitioning homomorphisms by the tuple of images $(h(x_1), \ldots, h(x_m))$ gives

$$\hom_{D(q^\circ)}(G, e) \;=\; \sum_{(e_1,\ldots,e_m):\, e_? = e} \;\prod_{\text{atoms } (x_i, r, x_j)} \mu_G(e_i, r, e_j),$$

where $\mu_G(a, r, b) = 1$ if $(a, r, b) \in E(G^+)$ and 0 otherwise. Because $D(q^\circ)$ encodes exactly the atoms, every summand is a product of indicators of atoms; hence

$$\mathbf{1}[e \in q^\circ(G)] \;=\; \mathbf{1}\big[\hom_{D(q^\circ)}(G, e) \geq 1\big]\,.$$

Set $\mathcal{W}_{q^\circ} = \{D(q^\circ)\}$ (a single witness of order m); membership is decided by one anchored hom-count.

Step 2 (disjunction, DNF). If $q = \bigvee_\ell c_\ell$ with conjunctive clauses c_ℓ, then $\mathbf{1}[e \in q(G)] = 1 - \prod_\ell (1 - \mathbf{1}[e \in c_\ell(G)])$, a polynomial in the clause indicators, each of which is Step-1 decidable. The witness family is the union of the clause databases.

Step 3 (atom-level negation, inclusion–exclusion). Let $q = c \wedge \neg a_1 \wedge \cdots \wedge \neg a_k$ where c is conjunctive and the a_i are negated atoms (possibly with fresh existential variables, as in `pin`/`pni`). Then

$$\mathbf{1}[e \in q(G)] = \mathbf{1}[e \in c(G)] \cdot \prod_{i=1}^{k} \big(1 - \mathbf{1}[e \in (c \wedge a_i)(G)]\big).$$

Expanding the product,

$$\mathbf{1}[e \in q(G)] = \sum_{S \subseteq [k]} (-1)^{|S|}\, \mathbf{1}\big[e \in (c \wedge \bigwedge_{i \in S} a_i)(G)\big],$$

and each $(c \wedge \bigwedge_{i \in S} a_i)$ is again a positive conjunctive query whose canonical database D_S is a witness of order at most $m + \sum_i |\mathrm{var}(a_i)|$. Hence

$$\mathbf{1}[e \in q(G)] \;=\; \sum_{S \subseteq [k]} (-1)^{|S|}\, \mathbf{1}\big[\hom_{D_S}(G, e) \geq 1\big],$$

a finite integer combination of anchored hom-counts; negated atoms contribute through inclusion–exclusion as claimed. Combining Steps 2 and 3 handles arbitrary DNF with negation. The bound on witness order is $|\mathrm{var}(q)| + \sum_i |\mathrm{var}(a_i)|$, finite and independent of G. □

Remark. The theorem states a sufficient statistic for *decidability*; executors in practice approximate thresholded counts by smooth scores. The smooth statement used in the main text—order-k executors factor through order-k signatures—follows because every feature consumed by such an executor is a function of the signature by Definition 3, and signatures of order k determine signatures of order $\leq k$.

III. Proof of Theorem 2 (Separation)

Theorem 2. *There is an explicit, linear-time constructible pair of query worlds (G_1, q_1), (G_2, q_2) such that (i) every score-factored fuzzy pipeline outputs identical rankings for q_1 in G_1 and q_2 in G_2, while (ii) their ground truths differ and an order-2 witness aggregator separates them.*

Proof. Construction. Entities: anchors a, targets t_1, t_2, t_3, t_4, marker hubs z. Relations: r_a; ambiguous relations r_1, r_2; type markers τ_1, τ_2. In each *block* b: (a) r_a-edges from the anchor a_b to all four targets; (b) a hidden half $H_b \subset \{t_{b1}, ..., t_{b4}\}$ with $|H_b| = 2$, chosen uniformly; (c) each target in the r_1-half carries a τ_1-edge and each target in the r_2-half a τ_2-edge; (d) in *demonstrator* blocks only, the local r_1/r_2 edges from the anchor are observed. The query in a hidden block is $q_b = (a_b, r_a) \wedge (a_b, r_q)$ with $r_q \in \{r_1, r_2\}$; its ground truth

is the two targets of the r_q-half, determined by the hidden edges—equivalently, by which targets carry τ_{r_q}-markers, since the construction ties the marker type to the half.

(i) Score-factored pipelines collide. Fix any executor whose projection returns per-entity scores and whose logical operators are pointwise functions of per-entity branch scores (products, probabilistic sums, complements, or any fixed continuous combinations thereof). In a hidden block the anchor has *no observed* r_q-edges; conditional on the block's local observations, the branch distribution over entities is a function of (a_b, r_q)'s local features only. By construction the local features of (a_b, r_q) are identical across the two worlds (both are "anchor with r_a-fanout four and unobserved r_q"), and they are identical for $r_q \in \{r_1, r_2\}$ within a world up to relabeling of relation symbols, which transferability (Definition 2) forces to act as an isomorphism. Hence the computed branch marginals, and therefore the final pointwise combinations and rankings, coincide across (G_1, q_1) and (G_2, q_2): the pipeline cannot be correct on both, because the ground truths differ (disjoint answer sets). Its expected MRR equals that of ranking four candidates of which two are relevant uniformly at random: $\frac{1}{4}(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}) \cdot 2 = \frac{2}{3}$ rounded to 0.67 (matches the empirical 0.673–0.703).

(ii) Order-2 aggregation separates. Define the aggregator that scores a target t in branch r_q by the indicator that t carries a τ_{r_q}-edge. This is a function of the order-2 signature: the pair (r_q, τ_{r_q}) co-occurrence is a two-node pattern count in the relation-interaction graph, and target markerhood is a one-hop anchored count. The aggregator's intersection then returns exactly the two relevant targets; MRR $= 1$. This is precisely the rule our order-2 gate variant implements (branch relation feature concatenated into the gate), and the empirical 1.000 matches. □

Remark (harmlessness in distribution). When the projection is trained to near-calibration on the query distribution, equal marginals co-occur with equal groundings, so the collision of (i) does not reduce accuracy—consistent with our paired-collision analysis on FB15k-237 (output Spearman 0.999, MRR ≈ 1). The separation bites exactly when local evidence is missing or shifted, which is the zero-shot regime of §6.3.

IV. Proof of Theorem 3 (Learnable-Operator Characterization)

Theorem 3. *The class $\mathcal{O}_k$ of order-k operators satisfying transferability is exactly the class of branch-permutation-invariant functions computable from multisets of order-k branch signatures. Fixed continuous t-norms/t-conorms act pointwise on order-0 evidence and form a strict subclass of measure zero inside any smoothly parameterized order-≥ 1 family containing them.*

Proof. (*Structure of the class.*) An operator ω consumes the branch outputs of a plan prefix. Transferability (Definition 2) implies ω is invariant under entity and relation relabelings; since the plan's branch list is unordered (conjunction and disjunction are commutative in the query DAG), ω is additionally invariant under branch permutations. By definition of aggregation order, ω is a measurable function of the multiset $\{s_i, \sigma_i\}_{i \leq m}$ of branch scores and order-k signatures. Conversely, any branch-symmetric measurable function of those multisets defines an element of $\mathcal{O}_k$: it consumes no identifiers, and multiset arguments make the relabeling action an isomorphism, so equivariance holds.

(*Fixed t-norms are order-0 and strict.*) A continuous t-norm combination applied pointwise over entities depends on the branch score vectors only—order-0 evidence—hence lands in $\mathcal{O}_0 \subsetneq \mathcal{O}_k$ for $k \geq 1$; strictness is witnessed by Theorem 2's separator, which is not expressible by any pointwise combination.

(*Measure zero.*) Let $\{T_\theta\}_{\theta \in \Theta}$ be a smoothly parameterized family containing the product at θ^*, e.g., the Aczél–Alsina family with $\theta^* = 1$, and let the gate be a smooth function of parameters and evidence. The set of parameters for which the operator equals T_{θ^*} pointwise is the preimage of a point under a smooth map into a space of higher dimension; by Sard's theorem it has measure zero. (For the one-parameter Aczél–Alsina family the preimage of $\{\theta^*\}$ under a non-degenerate gate is a codimension-one submanifold.) □

V. Lemma 1 (Exact Degeneration)

Lemma 1. *Setting every* SQA *gate output to the constant c with $\sigma(c) = \frac{1-\theta_0}{3.5}\big|_{\theta_0=1}$, i.e. $\theta \equiv 1$, makes the gated intersection equal to the product t-norm for all inputs, and dually for union.*

Proof. With $\theta_i \equiv 1$ for all branches,

$$T_\Theta(s_1, \ldots, s_m) = \exp\left[-\sum_i (-\ln s_i)\right] = \prod_i \exp(\ln s_i) = \prod_i s_i,$$

using $\sum_i (-\ln s_i) = -\ln \prod_i s_i$ and continuity of exp; the dual statement for union follows by applying the identity to complements $1 - s_i$. Numerically we verify $\max |T_\Theta - \prod s_i| < 10^{-3}$ at the forced gate on random inputs. Note the exponent $1/\theta_1 = 1$ as well, so no branch asymmetry survives. □

VI. Proposition (Transfer Continuity)

Proposition 1. *Define $d_{\hom}(G, G')$ as the ℓ_∞ discrepancy of anchored witness-count profiles over all patterns of order $\leq k$. For any order-k executor whose aggregation is L-Lipschitz in its inputs, $|F(G, q)(e) - F(G', q)(e')| \leq L\, d_{\hom}(G, G')$ for anchors e, e' corresponding under the profile.*

Proof sketch. The projection's messages are continuous functions of the anchored profiles (each hop composes counts along observed edges, Lipschitzly in the counts by the mean-value bound and the bounded degree of the plan). The order-k operator is L-Lipschitz in the branch scores and signatures by hypothesis. Chaining the two Lipschitz maps and taking the supremum over the finitely many order-$\leq k$ patterns yields the claim. The full multi-relational graphon formalism (existence of limits of anchored profiles, tightness) follows the classical bounded-degree convergence theory; we include it for completeness in the released technical note. □

VII. Experimental Details Referred to in the Main Text

A. Hyperparameters

Projection: 2-layer conditional propagation per hop, hidden 64. Relation encoder: 3-layer typed MP over G_R, hidden 64. Gates: 2-layer MLP ($65 \to 64 \to 1$; order-2 variant $129 \to 64 \to 1$), $\theta \in (0.5, 4)$ via $0.5 + 3.5\sigma(\cdot)$. Training: AdamW, ranking loss (pretraining, lr 10^{-3}, 50k steps, batch 12) or official balanced logit-BCE (fine-tuning, lr 5×10^{-4}, 10k steps, batch 32). Total executor parameters ≈ 145k (gated).

B. Inference cost

Measured on one RTX PRO 6000, FB15k-237 ($|V|$=14,505, $|E|$=544k with inverses): fixed operators 21.3 it/s (batch 12, all types); gated 19.6 it/s; gated order-2 19.1 it/s—an 8–10% overhead over fixed operators, negligible against projection cost. Zero-shot executor fusion (router) doubles projection cost (two executors), $\sim 2\times$.

C. Gate statistics after official fine-tuning

Empirical θ on sampled `2i` queries concentrates in $[0.52, 0.57]$—systematically *softer* than the product point $\theta = 1$: the learned optimum relaxes conjunction strength rather than sharpening it, consistent with the large disjunction gains (`2u` 0.963) under fuzzy margins.